Chapter 1

Sampling and Experimentation

1.1 Planning a Study

Every day, decisions get made based on data. A city council votes on a new transit route based on ridership surveys. A school principal adjusts the lunch schedule based on student feedback. A pharmaceutical company decides whether to release a new drug based on clinical trial results. In every case, the quality of the decision depends entirely on the quality of the data behind it.

To collect data well, you need a good plan! That plan must specify what question you are trying to answer, how you will gather information, and who you will gather this information from.

1.1.1 Vocabulary

Five terms form the foundation of everything that follows. Learn to distinguish them from each other, because you will probably be tested on this.

population The entire group of individuals or items that you want to learn about. The population is defined by the question you are asking.

frame Also called a sampling frame. A list of all the members of the population from which a sample will be drawn. The frame is usually close to the population, but the two are not always identical. A school's enrollment database, a city's voter registration list, and a company's customer records are all examples of frames.

sample The part of the population that is actually studied. You collect data from the sample and use it to draw conclusions about the population.

sample survey The process of collecting information from a sample. The information gathered is then used to make inferences about population parameters.

census The process of collecting information from every member of the population rather than just a sample. A census gives complete information, but it comes at a cost.

The relationship between these terms is worth pausing on. The population is the full group you care about. The frame is your working list of that group. The sample is the slice of the frame you actually study. A census is what happens when the sample and the population are the same thing.

Exercise 1 Identifying the Population and Sample

A city's parks department wants to understand how often residents use public parks. Staff randomly select 200 households from the city's water billing records and conduct phone interviews.

1. What is the population?
2. What is the frame?
3. What is the sample?
4. Is this a census? Explain.

Working Space

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1.1.2 Why Not Always Use a Census?

A census tells you about everyone in the population. That sounds great, but it is not always realistic. If the group is small, like a football team, you can measure everyone.

Most of the time, a census is too difficult because:

- The population is too large. A company that wants to know what all households watch cannot contact every single one. - The measurement destroys the item. If testing a battery kills it, checking every battery would waste them all. - The information gets old too fast. If a company waits months to ask everyone, the results may no longer match what people think.

Exercise 2 Census or Sample?

Working Space

For each situation, decide whether a census is feasible or whether a sample should be used. Briefly explain your reasoning.

Situation	**Census or Sample? Explain**
A school librarian wants to know which of her 8 student volunteers prefers morning or afternoon shifts.	
A battery manufacturer wants to test the lifespan of batteries by running them until they die.	
A government agency wants to estimate the average commute time of workers across a country of 40 million employed adults.	
A restaurant owner wants to know the dietary restrictions of the 22 guests attending a private event.	

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In most studies, sampling is the only practical option, and the quality of the conclusions you draw depends entirely on how well the sample was chosen.

1.2 Planning and Conducting Surveys

Not all samples are fair. How you pick people can make the group represent the population well, or it can leave out important parts.

1.2.1 Biased Sampling

A sample is *biased* if it favors some people or answers over others. Bias does not go away just because the sample is bigger.

These three common methods usually make biased samples.

Judgmental sampling is when one person decides who is included based on their own idea of what looks representative. A school inspector who visits only classrooms she already thinks are good is using judgmental sampling. The choice is based on opinion, not chance.

A *convenience sample* uses whoever or whatever is easiest to reach. A journalist who interviews only people at the bus stop outside his office is using a convenience sample. Those people might not represent all commuters.

A *voluntary response sample* happens when people choose whether to answer. An online poll on a news site will attract mostly visitors with strong opinions. People who do not care much usually do not respond.

All three skip chance. Sampling methods based on random selection give everyone a fair shot and usually make more trustworthy results.

1.2.2 Simple Random Sampling

A *simple random sample* (SRS) is a sample chosen so every group of the same size has the same chance of being picked.

There are two ways to do it.

With *sampling with replacement*, you put someone back in the pool after choosing them. They could be picked again.

With *sampling without replacement*, you do not put them back. Each person can only be chosen once.

To get an SRS, you need a real random method.

- Assign every member of the population a number. Use a random number table, entering at any random point and reading numbers sequentially. Select members whose numbers appear, skipping values outside your range and any repeats, until you have the sample size you need.
- Use a calculator. On a TI-84, `MATH` → `PRB` → `5:randInt(` generates random integers. The command `randInt(1,500,30)` produces 30 random integers between 1 and 500. Ignore repeats.
- Use a physical randomization device. Write each member's identifier on an identical slip of paper, mix thoroughly in a container, and draw without looking until the sample is complete.

The essential requirement in every case is that chance, not judgment or convenience, governs who is selected.

Exercise 3 Selecting an SRS

A regional transit authority wants to survey 25 of its 300 bus drivers about workplace safety conditions.

1. Describe a complete procedure for selecting an SRS of 25 drivers using a random number table.
2. A supervisor suggests selecting the 25 drivers who are easiest to reach during the next shift change. What type of sample is this, and what bias might it introduce?
3. A colleague suggests selecting 5 drivers at random from each of the authority's 5 depots. Is this an SRS? Explain.

Working Space

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1.2.3 Other Methods of Random Sampling

Simple random sampling is not always the most practical or most efficient approach. Three other probabilistic methods appear regularly on the AP exam.

In *systematic sampling*, you list the population, choose a random start, and then pick every kth person. For example, to sample 40 students from 800, start at a random number between 1 and 20, then take every 20th student. This is simple when you have a full list.

In *stratified random sampling*, the population is split into groups called *strata*. Each group should be similar inside, like students in the same grade or patients in the same ward. Then you take a random sample from each group. This makes sure every part of the population is represented.

In *cluster sampling*, the population is split into groups called *clusters*. Each cluster should be like a mini-version of the whole population. Then you randomly pick some clusters and study everyone inside them. This works well when people are spread out over a wide area.

Bonus method: *Snowball sampling* is a non-probabilistic method used when the population of interest is difficult or impossible to reach through conventional sampling. The researcher identifies a small number of initial participants who meet the study criteria, and then asks those participants to refer others they know who also qualify. Each wave of referrals generates the next, like a snowball rolling downhill and growing as it goes. It is not a commonly tested method on the AP exam, but it is worth knowing as an example of a non-probabilistic sampling technique that can be useful in certain contexts, such as studying hidden or hard-to-reach populations (e.g., people experiencing homelessness, members of stigmatized groups, or individuals with rare diseases).

Exercise 4 Identifying Sampling Methods

Working Space

Identify the sampling method used in each scenario: SRS, systematic, stratified, or cluster.

Scenario	**Method**
A supermarket chain selects 10 of its 80 stores at random and surveys every employee in those stores.	
A university randomly selects 50 students from each of its four faculties to participate in a wellbeing survey.	
A water utility selects a random starting meter on its numbered list and then tests every 25th meter for contamination.	
A film festival puts all 900 registered attendee names in a database and uses software to randomly select 90.	

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1.2.4 Bias in Surveys

Even a good random sample can still give bad results if the survey is set up badly. Bias can happen at many steps, not just when you pick people.

Sampling error is the normal difference between a sample and the population. Different samples can give different answers just by chance. This is not the same as bias. Sampling error gets smaller with a bigger sample.

The following types of bias are more serious because they do not shrink with larger samples. They push results systematically in a particular direction regardless of sample size.

Response bias occurs when respondents give answers that do not reflect their true beliefs or behaviors. This often happens when the topic is sensitive or when the presence of an interviewer creates social pressure. A survey on hand-washing habits conducted by a hygiene inspector at a food preparation facility will likely overstate how often workers wash their hands, because workers know what answer the inspector wants to hear.

Nonresponse bias occurs when individuals selected for the sample cannot be reached or choose not to participate, and those non-respondents differ systematically from those who do respond. A survey mailed to parents about school satisfaction will be completed at higher rates by engaged parents than by disengaged ones. The result overstates satisfaction among the parent community as a whole.

Undercoverage bias occurs when part of the population is excluded from the sampling frame entirely. A survey about internet access conducted through online banner advertisements cannot reach people who lack internet access. Those excluded are precisely the group most relevant to the question.

Wording effect bias occurs when the phrasing of a question influences the answer. Consider the difference between these two ways of asking essentially the same thing:

- ”Do you support stricter emissions standards for vehicles?”
- ”Do you support stricter emissions standards for vehicles, even if it means higher car prices and fewer model choices?”

The second version frames the question with costs attached, making a ”no” response far more likely even among people who genuinely support the policy. Neutral, direct wording is essential for unbiased responses.

Exercise 5 Diagnosing Bias

Identify the most likely type of bias in each scenario and explain why it applies.

1. A streaming platform sends a satisfaction survey to all subscribers but receives responses from only 8% of them. The platform reports that 91% of subscribers are satisfied.
2. A survey on recycling habits is conducted by going door-to-door in affluent neighborhoods only, because those streets are easier and safer to walk.
3. An interviewer asks employees: "Our new open-plan office is designed to encourage collaboration. How much has it improved your productivity?"
4. A survey on alcohol consumption is conducted at a community health fair by a team wearing hospital uniforms. Respondents significantly underreport their weekly intake.

Working Space

Answer on Page 14

1.3 Planning and Conducting Experiments

Surveys tell you what people think or report. Experiments tell you what actually causes what. The distinction determines what conclusions you are entitled to draw when the data comes in.

1.3.1 Experiments vs. Observational Studies

The defining question is whether the researcher assigns the conditions being studied or simply observes them as they exist.

In an *observational study*, the researcher collects data on subjects without intervening. The variables of interest already exist in the subjects; the researcher records what is there. A sociologist who tracks the grades and sleep hours of university students over a semester, without changing anyone's sleep schedule, is conducting an observational study.

In an *experiment*, the researcher imposes conditions on subjects and then measures the effect or change. The researcher creates the differences rather than observing them.

Why does the distinction matter? Only a properly designed experiment can support a cause-and-effect conclusion. An observational study can show that two variables are associated, but a third variable you did not account for could explain the relationship entirely. This uninvited third variable is called a *confounding variable*: one whose effect on the response cannot be separated from the effect of the variable you are actually studying.

In the observational sleep study, students who sleep more might also exercise more, eat better, or have lower course loads. Any of these could be responsible for better grades. There is no way to figure out which effects correlate to which behaviours.

There are situations where an experiment is unethical or impossible: you cannot randomly assign people to smoke for twenty years, or randomly select cities to experience floods. In these cases, observational data is all we have, and conclusions must be drawn with caution.

Exercise 6 Experiment or Observational Study?

For each study, identify whether it is an experiment or an observational study. Then state whether a cause-and-effect conclusion is justified and why.

Working Space

Study	Type	Causal conclusion?
Researchers randomly assign 180 seedlings to three groups and water each group with a different fertilizer solution. After eight weeks they measure plant height.		
A researcher records the screen time and attention span of 500 children aged 8 to 12, then looks for a relationship between the two.		
A company randomly selects half its customer service team to complete a new communication training program, then compares customer satisfaction scores between the two groups.		
A historian finds that cities with more libraries per capita had higher literacy rates in the early 20th century.		

Answer on Page 14

1.3.2 Vocabulary for Experiments

Experiments have their own vocabulary. These terms build on each other, so it helps to learn them in order.

Explanatory variable Also called the independent variable. The variable whose effect on the response is being studied. In a study of whether LoFi music affects study focus, the type of content being studied is the explanatory variable.

Response variable Also called the dependent variable. The outcome being measured. In a study of whether background music affects study typing speed, the typing speed is the response variable.

Experimental unit The smallest unit to which a treatment is applied. In a study of plants, each plant is an experimental unit. In a study of people, each person is an experimental unit.

Factor A variable whose effect on the response is of interest. Factors can be qualitative (type of music: classical, jazz, Afrobeat) or quantitative (temperature in degrees Kelvin).

Levels The specific values of a factor used in the experiment. If the factor is type of music with three options, there are three levels.

Treatment A specific combination of factor levels applied to experimental units. With one factor and three levels, there are three treatments. With two factors at two levels each, there are $2 \times 2 = 4$ treatments.

Control group A group that receives no treatment or a standard baseline condition. It provides a reference point for comparison with the treatment groups.

Placebo A dummy treatment that resembles the real treatment but contains no active component. Used to isolate the genuine effect of the treatment from the *placebo effect*: the tendency for people to show change simply because they believe they are receiving treatment.

1.3.3 The Three Principles of Experimental Design

A well-designed experiment rests on three principles.

Control means keeping everything the same for all groups except the treatment. It is not just having a control group. Control helps make sure the treatment is the only real difference.

Randomization means using chance to place experimental units into groups. It helps keep the groups fair and spreads unknown factors across all groups.

A *lurking variable* is something that can affect the response but is not part of the experiment.

Replication means including enough experimental units in each group so that chance variation among individuals does not swamp the treatment effect. If a new training method is tested on only two employees per group, a single unusually high or low performer could dominate the group's result. With 30 employees per group, individual extremes average out.

Exercise 7 Applying the Three Principles

Working Space

A horticulture researcher wants to test whether exposing seedlings to classical music for two hours per day increases their growth rate compared to no music exposure. She has 60 seedlings of the same species, grown from the same seed batch, available for the study.

1. Describe how randomization should be applied.
2. What does control mean in this context? List at least three variables that should be held constant.
3. The researcher initially plans to use 5 seedlings per group. A colleague recommends 30 per group. Which principle does this change address, and why?
4. Should there be a control group? What would it look like?

Answer on Page 15

1.3.4 Blinding

This is a draft chapter from the Kontinua Project. Please see our website (`https://kontinua.org/`) for more details.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 2)

1. The population is all households in the city.
2. The frame is the city's water billing records.
3. The sample is the 200 households selected and interviewed.
4. No. A census would require interviewing every household in the city. Only 200 were selected here.

Answer to Exercise 2 (on page 3)

- **Library volunteers:** Census. With only 8 volunteers, asking all of them takes almost no effort.
- **Battery lifespan:** Sample. Testing can destroy the batteries, so a census would leave nothing to sell.
- **Commute times:** Sample. The population of 40 million is far too large for a full census to be practical.
- **Dietary restrictions:** Census. With only 22 guests, collecting complete information is straightforward and important for planning.

Answer to Exercise 3 (on page 5)

1. Number the 300 drivers from 001 to 300. Enter a random number table at any point and read three-digit numbers. Select any driver whose number falls between 001 and 300, skipping values from 301 to 999 and any numbers that have already been selected, until 25 drivers have been chosen. Alternatively, use `randInt(1,300,25)` on a TI-84, ignoring repeats.
2. This is a convenience sample. Drivers who are easy to reach at shift changes may work particular routes or shifts that differ systematically from others. For example,

they may have different safety exposure than drivers on overnight or rural routes.

3. No. An SRS requires that every possible group of 25 drivers from all 300 has an equal chance of being chosen. This procedure cannot produce a sample where all 25 come from the same depot, so not all groups of 25 are possible. This is a stratified sample, not an SRS.

Answer to Exercise 4 (on page 6)

- **Supermarket:** Cluster sampling. The stores are clusters, which means entire clusters are included once selected.
- **University:** Stratified random sampling. The four faculties are strata. An SRS is taken from each.
- **Water utility:** Systematic sampling. Random start followed by every 25th unit.
- **Film festival:** Simple random sample. Every possible group of 90 attendees is equally likely.

Answer to Exercise 5 (on page 8)

1. Nonresponse bias. The 92% who did not respond are unlikely to be satisfied in the same proportions as those who did. Highly satisfied and highly dissatisfied subscribers are both more motivated to respond than those with neutral feelings, but the pattern of non-response is unknown and the reported figure cannot be trusted.
2. Undercoverage bias. Residents of lower-income neighborhoods are excluded from the frame entirely. Recycling behavior may differ systematically by neighborhood type, so the results cannot generalize to the full population.
3. Wording effect bias. The question assumes the office has improved productivity and asks only ”how much.” Respondents are not given a neutral opportunity to say productivity has not changed or has declined.
4. Response bias. The formal appearance of the interviewers creates social pressure to give a health-conscious answer. Respondents adjust their reported behavior to match what they believe the interviewers expect.

Answer to Exercise 6 (on page 10)

- **Seedlings:** Experiment. Fertilizer was randomly assigned. Causal conclusion is justified.
- **Screen time and attention:** Observational study. No assignment was made. Causal conclusion is not justified; many confounding variables (parenting style, diet, sleep) could explain the relationship.
- **Customer service training:** Experiment. The training was randomly assigned. Causal conclusion is justified.
- **Libraries and literacy:** Observational study. No assignment was made. Causal conclusion is not justified; wealthier cities may have had both more libraries and higher literacy for unrelated reasons.

Answer to Exercise 7 (on page 12)

1. Number the 60 seedlings from 1 to 60. Use a random number generator to select 30 for the music group. The remaining 30 form the no-music group.
2. Control means growing all seedlings under identical conditions except for the music. Variables to hold constant include: soil type and quantity, pot size, amount and frequency of watering, light exposure, temperature, and humidity.
3. Replication. With only 5 seedlings per group, a single unusually fast or slow grower could distort the group average. With 30 per group, individual variation averages out and a real effect of the music is more detectable.
4. Yes. The no-music group serves as the control group, establishing the baseline growth rate without the treatment. Without it, there is nothing to compare the music group's growth against.

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